Grand Quiz Spring 2021

Subject Code PHY 101 lecture 1 to 22

Solved By Riz Mughal

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I'm providing 100% correct quiz solution. You can visit my YouTube channel and get more information about all other subjects' quizzes and final year project (CS619).

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RIZ MUGHAL (SQA ENGINEER)

PHY101:Grand Quiz

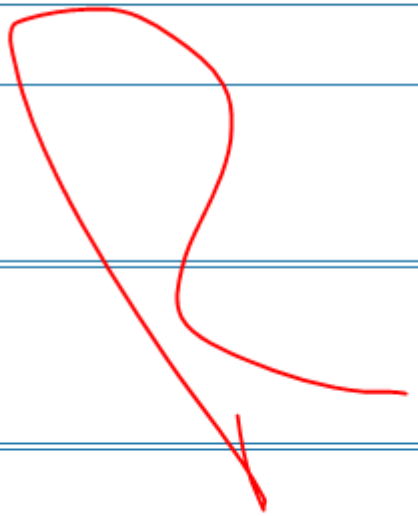
Quiz Start Tim

Question # 1 of 30 (Start time: 09:40:17 AM, 30 June 2021)

A fire whistle emits a tone of 170 Hz. Take the speed of sound in air to be 340m/s. The wavelength of this sound is about:

Select the correct option

- (●) 2.0m
- ( ) 3.0m
- ( ) 0.5m
- ( ) 1.0m



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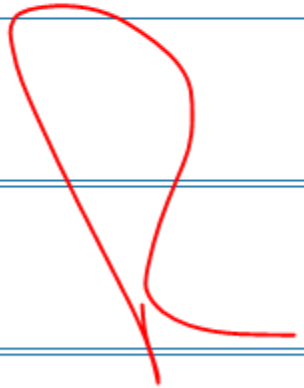
Question # 2 of 30 (Start time: 09:40:36 AM, 30 June 2021)

The angular momentum L of a rigid body is given by:

Select the correct option

- (●) Iw
- ( ) rw
- ( ) mrI
- ( ) mrw



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Quiz Start Time: 09:40 AM, 30 June

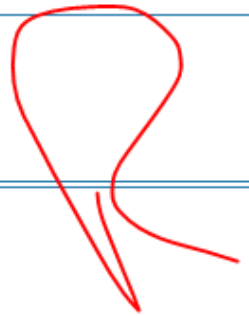
Question # 3 of 30 (Start time: 09:40:53 AM, 30 June 2021)

Total Ma

A water bed that is 1.5 m wide and 2.5 m long weighs 1055 N. Assuming the entire lower surface of the bed is in contact with the floor, what is the pressure the bed exerts on the floor?

Select the correct option

- ○ 270 Pa
- ○ 250 Pa
- ● 280 Pa
- 260 Pa



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Question # 4 of 30 (Start time: 09:41:08 AM, 30 June 2021)

A particle oscillating in simple harmonic motion is:

Select the correct option

- never in equilibrium because it is in motion
- in equilibrium at the ends of its path because its velocity is zero there

in equilibrium at the ends of its path because its velocity is zero there

- never in equilibrium because there is always a force
- in equilibrium at the center of its path because the acceleration is zero there

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Question # 5 of 30 (Start time: 09:41:35 AM, 30 June 2021)

Young's modulus can be used to calculate the strain for a stress that is:

Select the correct option

- (●) well below the yield strength
- ( ) well above the yield strength
- ( ) just below the ultimate strength
- ( ) just above the ultimate strength

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Quiz Start Time: 09:40 AM, 30 June

Question # 6 of 30 (Start time: 09:41:49 AM, 30 June 2021)

Total Ma

A mosquito's buzz is often rated with a decibel rating of 40 dB. Normal conversation is often rated at 60 dB. How many times more intense is normal conversation compared to a mosquito's buzz?

Select the correct option

- (●) 100
- ( ) 20
- ( ) 400
- ( ) 2

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Question # 7 of 30 (Start time: 09:42:03 AM, 30 June 2021)

Which of the following statement/s is / are true?

i. The weight of a man on the Moon is smaller than that on the Earth.
ii. The mass of a man is the same on both the Moon and the Earth.
iii. We cannot determine our own mass in the outer space because there is no gravity.

Select the correct option

- ○ (i) only
- ● (i) and (ii) only
- ○ (iii) only
- ○ (ii) and (iii) only

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Question # 8 of 30 (Start time: 09:42:20 AM, 30 June 2021)

The angular momentum vector of Earth about its rotation axis, due to its daily rotation, is directed:

Select the correct option

- ○ tangent to the equator toward the east
- ○ tangent to the equator toward the west
- ● north
- ○ south

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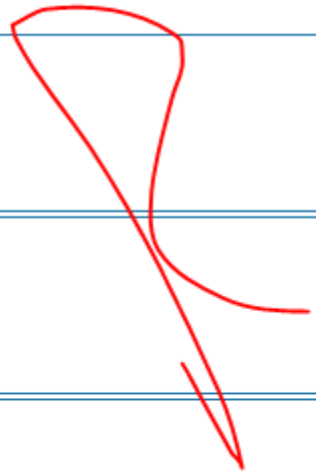
Question # 9 of 30 (Start time: 09:42:36 AM, 30 June 2021)

When body moves with constant acceleration the velocity time graph is:

Select the correct option

- ○ Hyperbola
- ● Straight line
- ○ Curve
- ○ Parabola



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Quiz Start Time: 09:40 AM, 30 Jun

Question # 10 of 30 (Start time: 09:42:51 AM, 30 June 2021) **Total M**

The persistence of audible sound due to the successive reflections from the surrounding objects even after the source has stopped to produce that sound is called

Select the correct option

- (●) Reverberation
- ( ) rarefaction
- ( ) Reflection
- ( ) Echo

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Quiz Sta

Question # 11 of 30 (Start time: 09:43:09 AM, 30 June 2021)

A vector A is added to a vector B. The resultant vector A + B have greatest magnitude when:

Select the correct option

- ○ The magnitude of vector A + B does not depend on the directions of A and B
- ○ When vectors A and B are perpendicular
- ● When vectors A and B are parallel and in the same direction
- ○ When vectors A and B are parallel and in the opposite direction

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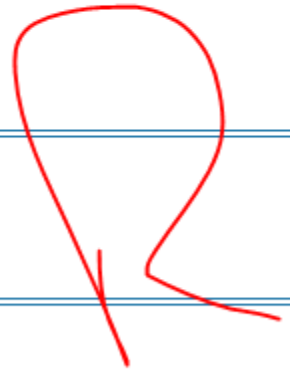
Question # 12 of 30 (Start time: 09:43:24 AM, 30 June 2021)

The unit of intensity of sound is:

Select the correct option

- ○ Candela
- ○ Decibel
- ● Kelvin
- ○ Meter



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Quiz Start Time: 09:40 AM, 30 June 2

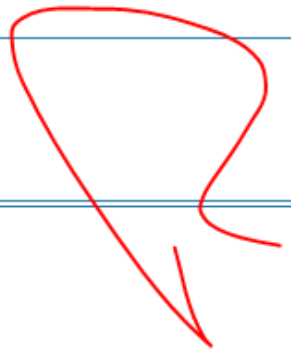
Question # 13 of 30 (Start time: 09:43:38 AM, 30 June 2021)

Total Mar

When number of bodies are such that they can exert force upon one another and no external agency exerts a force on them, they are said to form:

Select the correct option

- ○ Non-inertial frame of reference
- ○ Non isolated system
- ● An isolated system
- ○ An inertial frame of reference



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Question # 14 of 30 (Start time: 09:43:54 AM, 30 June 2021)

Because a buoyant force acts in the opposite direction of gravity,

Select the correct option

- ○ objects submerged in water have a net force larger than their weight.
- ● objects submerged in water have a net force smaller than their weight.
- ○ objects submerged in water appear to weigh more than they do in air.
- ○ objects submerged in water have a net force equal to their weight.

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Question # 15 of 30 (Start time: 09:44:08 AM, 30 June 2021)

If the mass of moving objects is doubled then its K.E becomes:

Select the correct option

- 4 times
- 5 times
- 16 times
- 2 times

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Question # 16 of 30 (Start time: 09:44:24 AM, 30 June 2021)

acceleration of an object must be zero at a point where:

Select the correct option

○	the average velocity is zero
●	the instantaneous velocity is constant
○	the instantaneous velocity is not zero but changing
○	the instantaneous velocity is zero but changing

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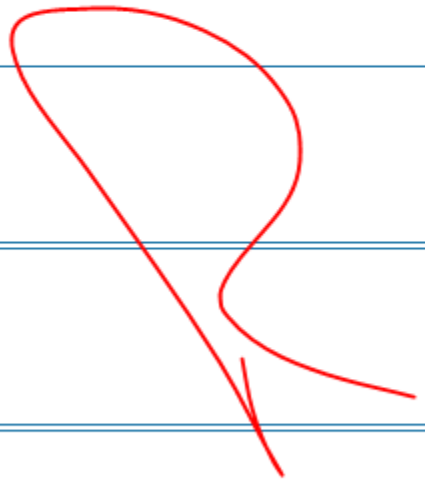
Question # 17 of 30 (Start time: 09:44:41 AM, 30 June 2021)

back and forth in a leftward and rightward direction. This type of wave is known as a ____.

Select the correct option

- ○ Mechanical
- ● Longitudinal
- ○ Transverse
- ○ Electromagnetic



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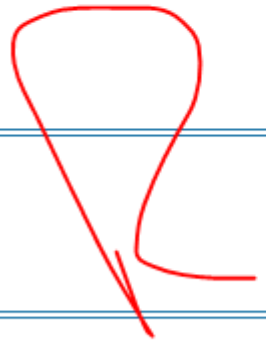
Question # 18 of 30 (Start time: 09:45:10 AM, 30 June 2021)

Final take off velocity of an airplane is 67 m/s. The length of runway is 2 km, the constant acceleration is:

Select the correct option

- 2.24 ms^-1
- 2.24 ms^-2
- 3.24 ms^-2
- 3.24 ms^-1



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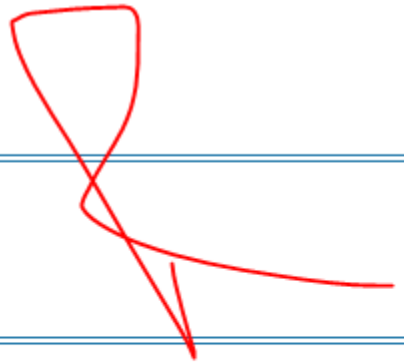
Question # 19 of 30 (Start time: 09:45:27 AM, 30 June 2021)

The frequency which is not audible to the human ear is

Select the correct option

- 50000 Hz
- 500 Hz
- 50 Hz
- 5000 Hz



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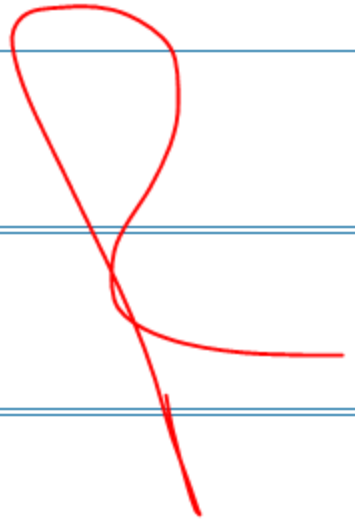
Question # 20 of 30 (Start time: 09:45:47 AM, 30 June 2021)

The speed of sound in medium depends upon

Select the correct option

- (●) properties of the medium
- ( ) amplitude
- ( ) frequency
- ( ) wavelength



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Question # 21 of 30 (Start time: 09:46:02 AM, 30 June 2021)

If a body whose mass is much less than a body at rest collide with it elastically, then it bounces back with:

Select the correct option

- ○ Half of the velocity
- ○ Double velocity
- ○ None of these
- ● Same velocity

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Question # 22 of 30 (Start time: 09:46:19 AM, 30 June 2021)

SI Unit of time period is _____.

Select the correct option

- nanosecond
- minute
- hour
- second

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Question # 23 of 30 (Start time: 09:46:36 AM, 30 June 2021)

Which of the following is not a characteristic of mechanical waves?

Select the correct option

- (●) They travel in a direction which is at right angles to the direction of the particles of the medium.
- ( ) They transport energy.
- ( ) They are created by a vibrating source.
- ( ) They consist of disturbances or oscillations of a medium.

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Question # 24 of 30 (Start time: 09:46:51 AM, 30 June 2021)

As the wavelength of a wave in a uniform medium increases, its frequency will _____.

Select the correct option

○	Remain the same
○	None of these
○	Increase
●	Decrease

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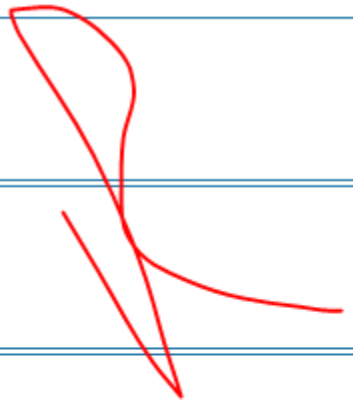
Quiz Start Tim

Question # 25 of 30 (Start time: 09:47:06 AM, 30 June 2021)

Which of the following has the smallest moment of inertia about the central axis if all have equal masses and radii?

Select the correct option

- ○ Ring
- ○ Disc
- ● Sphere
- ○ Spherical shell



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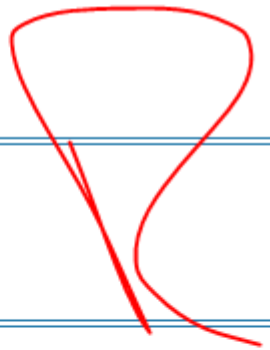
Question # 26 of 30 (Start time: 09:47:22 AM, 30 June 2021)

Which of the following cases is/are NOT a uniformly accelerated motion?
i.A feather falls from a certain height inside a vacuum tube.
ii.A ball rolls along a frictionless plane at uniform speed.
iii.A coin falls from a certain height in air but air resistance is negligible.

Select the correct option

- (i) and (ii) only
- (ii) only
- (ii) and (iii) only
- (i) only



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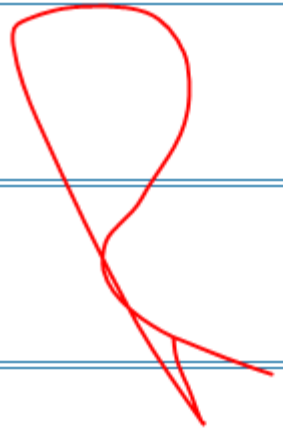
Question # 27 of 30 (Start time: 09:47:38 AM, 30 June 2021)

Before the density of an object can be found, what two measurements are required?

Select the correct option

- (●) mass and volume
- ( ) temperature and volume
- ( ) mass and length
- ( ) volume and length



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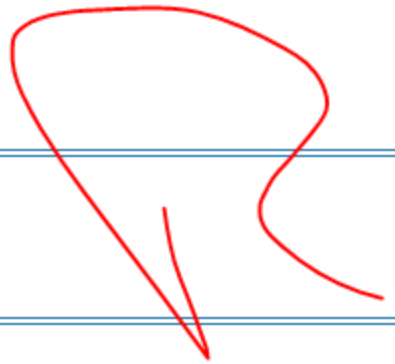
Question # 28 of 30 (Start time: 09:47:51 AM, 30 June 2021)

Distance covered during one vibration of an oscillating body in terms of amplitude A:

Select the correct option

- (●) 4A
- ( ) 2A
- ( ) A/2
- ( ) A



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Quiz Start T

Question # 29 of 30 (Start time: 09:48:06 AM, 30 June 2021)

If the distance between all pairs of particles of the body do not change by applying a force then the body is said to be

Select the correct option

- large
- flexible
- rigid
- small

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Quiz Start Time: 09:40 AM, 30 Ju

Question # 30 of 30 (Start time: 09:48:21 AM, 30 June 2021) **Total**

Two sound waves are traveling through a container of unknown gas. Wave A has a wavelength of 1.2 m. Wave B has a wavelength of 3.6 m. Th
frequency of wave B must be _____ the frequency of wave A.

Select the correct option

- ○ Three times larger than
- ○ The same as
- ● One-third
- ○ One-ninth

Thank you for watching

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